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PERSONAL BIO

Senior Data Engineer & AI Solutions Architect with 5+ years experience delivering end-to-end data platforms and intelligent applications across fintech, SaaS, and industrial analytics. Expert in designing scalable data pipelines, building AI-powered systems and Agentic AI with LLMs and machine learning models, and deploying robust cloud-native solutions on AWS, Azure, and GCP, serving millions of users.

WORK EXPERIENCE

Senior Data Engineer & AI Solutions Architect

04/2024 – present | Netherlands

Helixiora

- **Client Delivery & Solution Leadership:** Led customer-facing project delivery by partnering with stakeholders to design and deploy innovative, end-to-end solutions across Azure and AWS, translating business problems into scalable data, AI, and platform architectures.
- **Data Engineering:** Designed and built large-scale Azure Databricks pipelines using PySpark and SQL to ingest, clean, and transform multi-source datasets (Blob, CosmosDB); implemented real-time streaming pipelines with Kafka; and delivered curated Delta Lake layers aligned with Kimball-style dimensional modeling and data warehouse best practices to power analytics and ML workloads.
- **Full-Stack SaaS:** Led the end-to-end development and deployment of multiple production-grade AI SaaS platforms, serving millions of users. Built full-stack solutions using FastAPI, Next.js, and React Native with Expo for Android and iOS, integrating OAuth 2.0 authentication, PostgreSQL, and Stripe to deliver secure, scalable, cloud-native products.
- **AI Agents:** Architected and deployed multi-stage agentic RAG, Chatbot and automation systems using OpenAI Agent SDK, LangChain, LangGraph, CrewAI, Pinecone, and custom Python pipelines (Slack, Google Drive), enabling accurate context retrieval and intelligent workflow automation.
- **Machine Learning Engineering:** Built feature-ready datasets in Delta Lake, supported downstream ML model training, and optimized vector search & reranking pipelines, accelerating model iteration cycles and improving retrieval accuracy across enterprise LLM solutions.
- **LLM Fine-Tuning:** Fine-tuned domain-specific LLMs using Unsloth for specialized business use cases, optimized model performance with LoRA, and deployed the resulting fine-tuned models into production AI workflows, significantly improving inference accuracy.
- **LLM Inference & Deployment:** Deployed vLLM and llama.cpp for high-performance LLM serving, optimized for low latency and high-concurrency workloads.
- **Infrastructure, DevOps & Security:** Designed and automated CI/CD pipelines deploying containerized applications to AWS ECS using Docker, Terraform, and GitHub Actions; strengthened platform reliability with load balancing and scalable microservices; and implemented Zero-Trust controls, SSL/TLS hardening, firewall rules, and secure API gateways to ensure robust authentication, authorization, and data protection across SaaS and data platforms.
- **Monitoring & Observability:** Set up centralized logging, metrics, and dashboards using Prometheus and Grafana, enabling real-time performance monitoring, alerting, and proactive issue resolution across production environments.
- **Team Leadership & Mentorship:** Led and mentored junior and mid-level engineers through code reviews, architectural guidance, use of AI tools, and best-practice coaching, raising team capability and ensuring high quality engineering delivery.

Senior Data Engineer

02/2022 – 03/2024

Vaultoro

Manchester, United Kingdom

- Architected a **Kimball-style star-schema data warehouse** using **Elasticsearch** and **BigQuery**, enabling real-time KPI **dashboards** in Kibana and supporting data-driven decision making.

- Led the design and implementation of multiple **scalable ETL pipelines** with Python, **GCP Dataflow**, Scrapy, and managed workflow orchestration with Apache **Airflow**, processing over **10 million financial data rows daily** for real-time analytics.
- Co-led the development of a secure and scalable **FastAPI**-based Savings Platform, delivered in 3 months and now processing **\$2M+** in monthly customer deposits..
- Managed and optimized databases including **MongoDB, PostgreSQL, Elasticsearch, and BigQuery** by tuning queries, indexes, and partitions, resulting in significant performance improvements.
- Conducted deep data analysis with **Pandas** to detect anomalies and identify potential fraud patterns, enhancing platform security.
- Automated data validation workflows using Python scripts, ensuring pipeline integrity and achieving **100%** uptime for critical microservices.
- Deployed containerized applications with **Docker and Kubernetes**, ensuring high availability, scalability, and **streamlined CI/CD** operations.
- Developed and implemented a **Random Forest classification** model to identify high-value clients during signup, enabling personalized onboarding experiences.

Data Analyst

06/2019 – 10/2020 | Guwahati, India

SoftCrop IT

- Delivered actionable insights via **Tableau dashboards (waterfall/cohort analysis)**, improving stakeholder decision-making.
- Automated competitor **data scraping (Scrapy)** and ETL into **MySQL**, reducing manual labour.
- Analyzed sales and **geographic data** to guide strategic fibre network expansion.

EDUCATION

MSc Business Analytics and Big Data - University of Liverpool

11/2020 – 02/2022 | Liverpool, United Kingdom

Result: Distinction

Bachelor of Business Administration: Asian Institute of Management and Technology

08/2016 – 06/2019 | Guwahati, India

Result: First-Class

CORE SKILL

Consulting & Professional Skills: Customer Delivery, Stakeholder Management, Agile Collaboration, Technical Communication, Solution Design, Technical Roadmapping

Data Engineering: PySpark, SQL, Databricks, Airflow, Kafka, Delta Lake, Data warehousing, Data modeling

AI/ML: Scikit, Pytorch, RAG, LangChain, LangGraph, CrewAI, Unsloth, MLflow, Pinecone, Weaviate

Backend & Full-Stack: FastAPI, Django, NextJS, Expo, React, Go, C++

Database: PostgreSQL, MongoDB, Elasticsearch, BigQuery

DevOps: Docker, Kubernetes, Terraform, Git, GitHub Actions, CI/CD, Prometheus, Grafana

Cloud: Azure, AWS, GCP, VPS (DigitalOcean / Hetzner)

CERTIFICATES

IBM Data Science Professional Certificate	Microsoft Certified: Azure Fundamentals AZ-900	Microsoft Certified: Azure AI Fundamentals AI-900
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